

Trainings ▾

General Specifications

Listed below are general specifications for the engine. See each system section for additional specifications.

No injector adjustments are required for engines with electronic injectors.

QSK45 Engines

Horsepower	See the engine dataplate
Engine Speed	See the fuel system calibration data for speed rating
Aspiration	Turbocharged and Aftercooled
Bore and Stroke	159 mm [6.25 in] x 190 mm [7.48 in]
Displacement	45 liters [2746 C.I.D.]
Firing Order	1R-6L-5R-2L-3R-4L-6R-1L-2R-5L-4R-3L]
Engine Type	4 cycle, 60-degree vee, 12 cylinder

Compression Ratio

Industrial	Application-Dependent
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Two-Pump and Two-Loop Cooling System:

at 1800 rpm	14.5:1
at 1900 rpm	16.5:1
Power-Generation	Application-dependent

Jacket Water Aftercooled, Two-Pump and Two-Loop Cooling System, TA Luft:

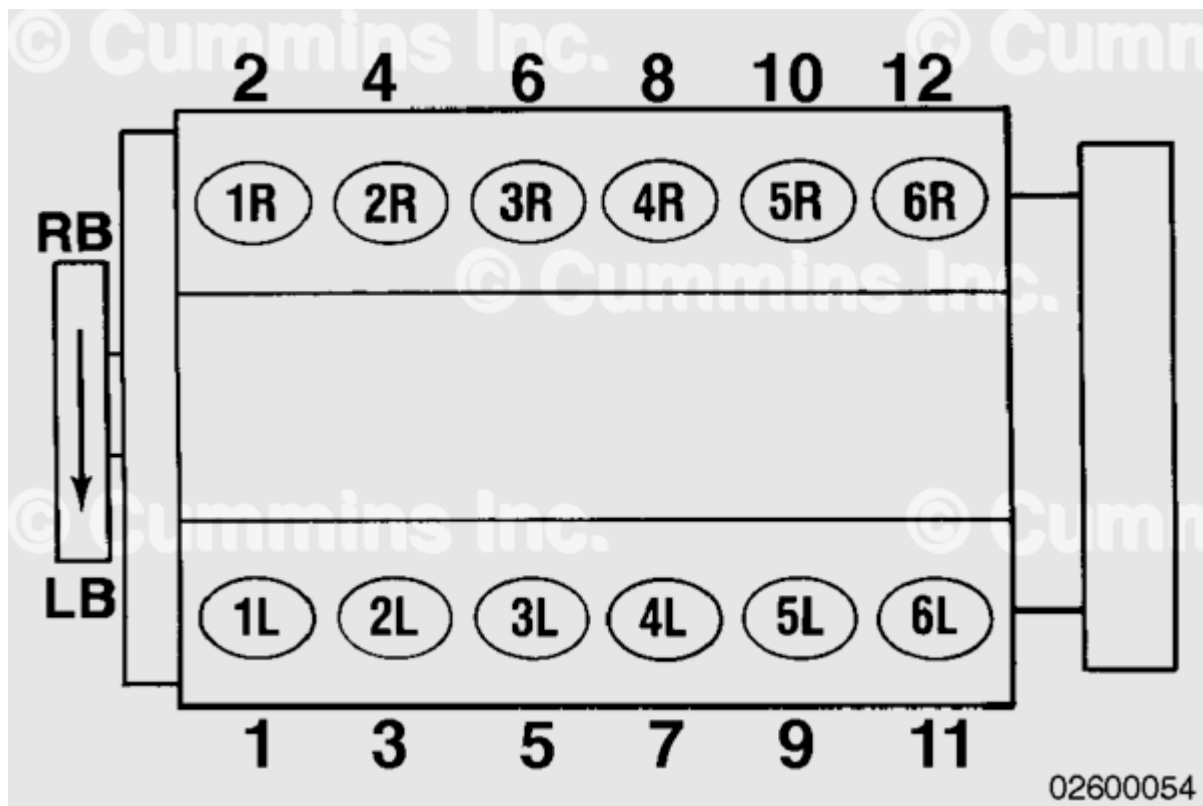
at 1800 rpm/60 Hz	14.5:1
at 1900 rpm/50 Hz	16.2:1

Engine Weight (estimates only):

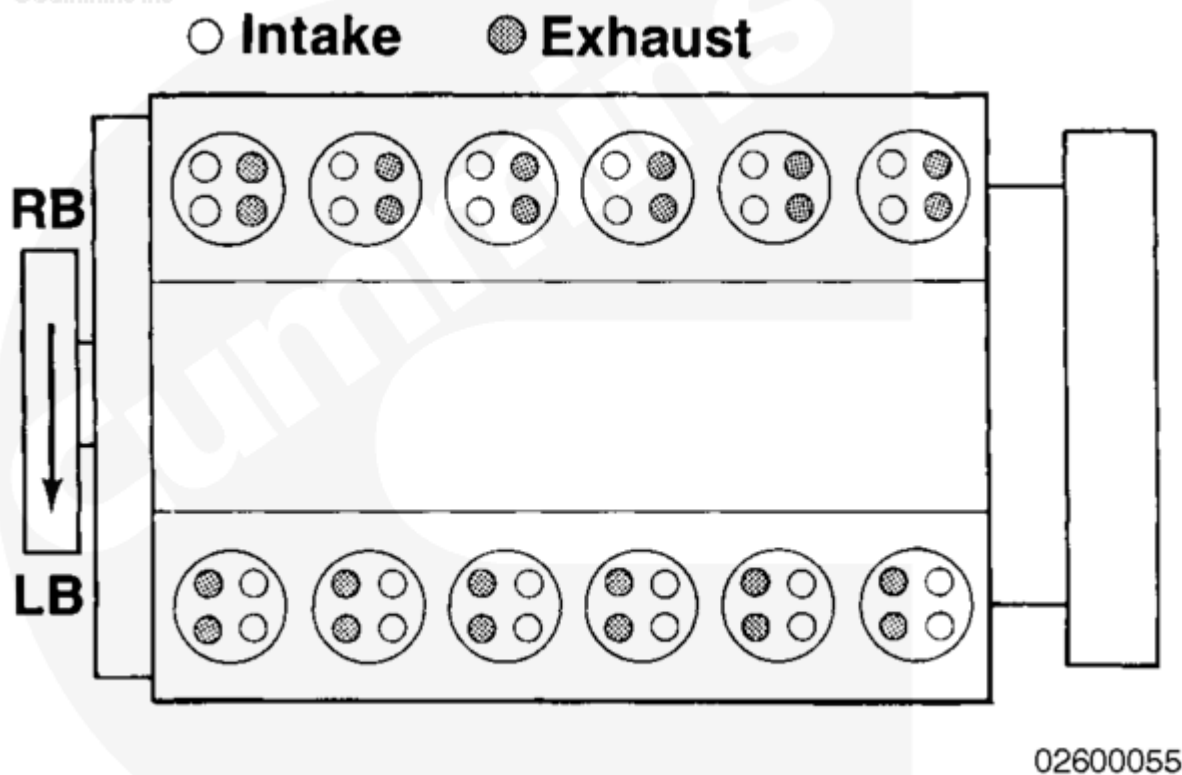
Wet	6123 kg [13.499 lb]
Dry	5813 kg [12,815 lb]
Crankshaft Rotation (viewed from the front of the engine)	Clockwise

Valve and Injector Settings:

Intake Valve Adjustment	0.36 mm [0.014 in]
Intake Valve Limits	0.28 to 0.43 mm [0.011 to 0.017 in]
Exhaust Valve Adjustment	0.81 mm [0.032 in]
Injector Outer Base Circle Method Adjustment (in engine)	19 N•m [168 in-lb]



Cylinder Numbering Sequence: RB = Right-bank of cylinders and LB = Left-bank of cylinders.



The intake and exhaust valve locations as shown in the illustration.

QSK60 Engines

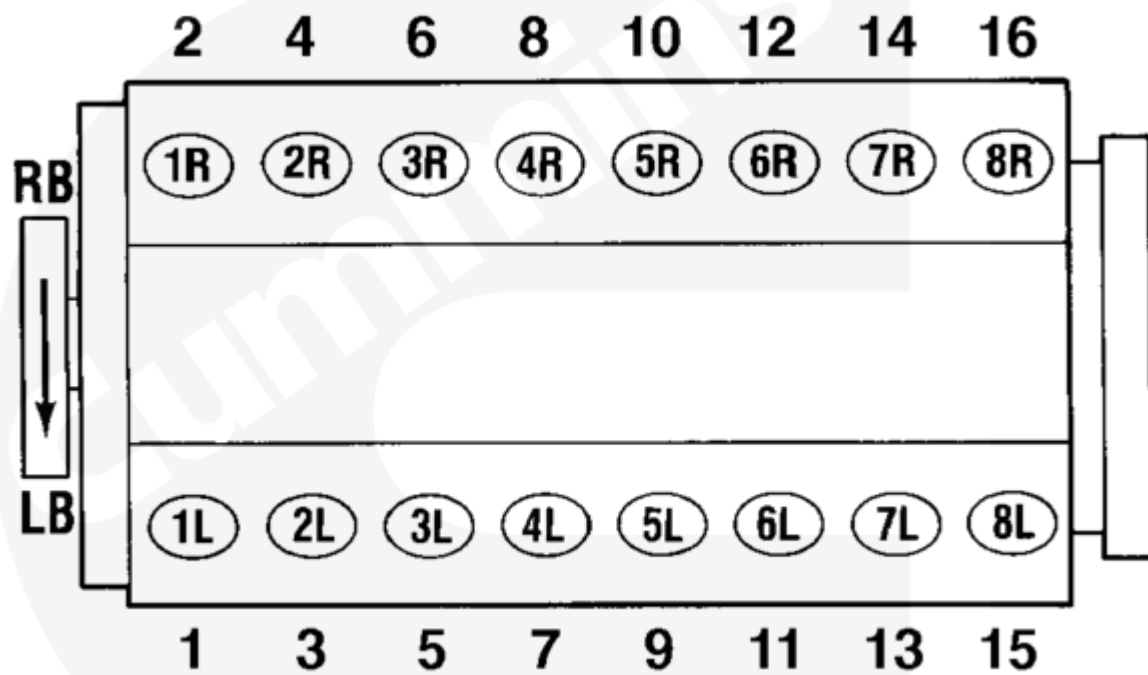
Horsepower	See the engine dataplate
Engine Speed	See the fuel system calibration data for speed rating
Aspiration:	
Single-Stage	Turbocharged and aftercooled
Two-Stage	Turbocharged, aftercooled, and intercooled
Bore and Stroke	159 mm [6.25 in] x 190 mm [7.48 in]
Displacement	60 liters [3660 C.I.D.]
Engine Type	4 cycle, 60-degree vee, 16 cylinder
Firing Order	1R-1L-3R-3L-2R-2L-5R-4L-8R-8L-6R-6L-7R-7L-4R-5L
Crankshaft Rotation (viewed from the front of the engine)	Clockwise

Compression Ratio with mechanically actuated injectors:

Marine	14.5:1
Industrial	Application-Dependent
Two-Pump and Two-Loop Cooling System:	
at 1800 rpm	14.5:1
at 1900 rpm	16.5:1
Power-Generation	Application-dependent
Jacket Water Aftercooled, Two-Pump and Two-Loop Cooling System, TA Luft:	
at 1800 rpm/60 Hz	14.5:1
at 1900 rpm/50 Hz	16.2:1
Compression Ratio With Electronically Actuated Injectors:	
All Applications	14.5:1
Industrial Single-Stage Engine Weight With Mechanically Actuated Injectors Full Length Sump (estimates only):	
Wet	8475 kg [18,685 lb]
Dry	7990 kg [17,614 lb]
Power Generation Single-Stage Engine Weight with Mechanically actuated Injectors and Full Length Sump (estimates only):	
Wet	8065 kg [17,781 lb]
Dry	7580 kg [16,711 lb]
Industrial Single-Stage Engine Weight with Electronically Actuated Injectors and Full Length Sump (estimates only):	
Wet	8745 kg [19,280 lb]
Dry	8260 kg [18,210 lb]
Power Generation Single-Stage Engine Weight with Electronically Actuated Injectors and Full Length Sump (estimates only):	
Wet	8205 kg [18,089 lb]
Dry	7720 kg [17,019 lb]
Industrial Two-Stage Engine Weight With Mechanically Actuated Injectors and Full Length Sump (estimates only):	
Wet	9518 kg [20,984 lb]

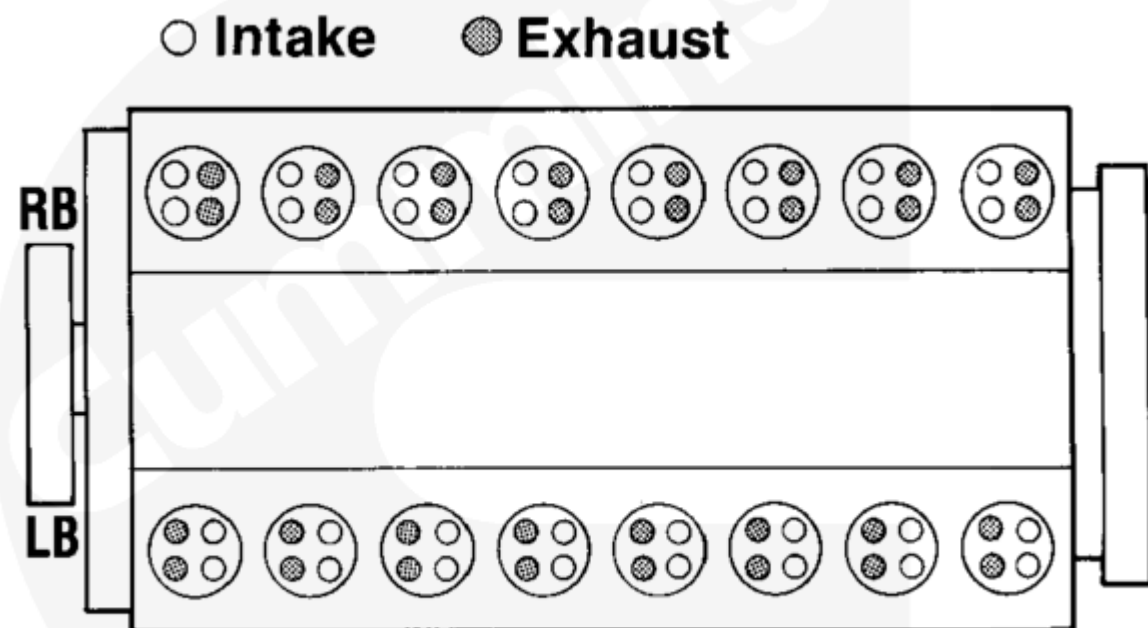
Dry	9000 kg [19,842 lb]
Industrial Two-Stage Engine Weight With Electronically Actuated Injectors and Full Length Sump (estimates only):	
Wet	9568 kg [21,095 lb]
Dry	9050 kg [19,952 lb]
Marine Engine Weight with Mechanically Actuated Injectors; Engine Only (estimate)*:	
Wet	9341 kg [20,595 lb]
Dry	8754 kg [19,299 lb]
Marine Engine With Heat Exchanger and Expansion Tank With Mechanically Actuated Injectors (estimate)*	
Wet	9817 kg [21, 643 lb]
Dry	9136 kg [20141 lb]
Marine Engine Weight With Electronically Actuated Injectors; Engine Only (estimate)*:	
Wet	9228 kg [20,343 lb]
Dry	8640 kg [19,048 lb]
Marine Engine With Heat Exchanger and Expansion Tank With Electronically Actuated Injectors (estimate)*	
Wet	9435 kg [20,801 lb]
Dry	8754 kg [19,300 lb]
Marine Engine Dimensions *	
Engine height	2420 mm [95.3 in]
Engine with heat exchanger and expansion tank	2420 mm [95.3 in]
Engine width	1730 mm [68.1 in]
Engine with heat exchanger and expansion tank	1730 mm [68.1 in]
Engine length	3357 mm [132.2 in]
Engine with heat exchanger and expansion tank	3602 mm [141.8 in]
Continuous	

2000 hp	1800 rpm
2200 hp	1800 rpm
Heavy Duty	
2300 hp	1900 rpm
2500 hp	1900 rpm
Prime power	
1563 kW at 50 Hz	1500 rpm
1900 kW at 60 Hz	1800 rpm
Crankshaft Rotation (viewed from the front of the engine)	Clockwise
Valve and Injector Settings:	
Intake Valve Adjustment	0.36 mm [0.014 in]
Intake Valve Limits	0.28 to 0.43 mm [0.011 to 0.017 in]
Exhaust Valve Adjustment	0.81 mm [0.032 in]
Exhaust Valve Limits	0.74 to 0.89 mm [0.029 to 0.035 in]
Injector Outer Base Circle Method Adjustment (in engine).	19 N•m [168 in-lb]
Note : No adjustment required for engines with electronically actuated injectors.	
* QSK60M-specific	



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Cylinder Numbering Sequence: RB = Right bank of cylinders and LB = Left bank of cylinders.



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Intake and exhaust valve locations.

